Predicting Remote Work Burnout & Social Isolation

Using Machine Learning to Flag At-Risk Employees Early

Team Members

Name	Role & Responsibilities
Eman Sarhan	Project Manager — coordinates tasks, tracks timeline, ensures deliverables are on time
Yasmeen Nofal	Data Cleaning — handles missing values, encodes categories, normalizes features
Joud Hijaz	EDA — creates visualizations, detects outliers, analyzes feature correlations
Nada Tabib	ML Modeling — trains and tunes Logistic Regression, Decision Tree, Random Forest
Maymona Khanfar	Evaluation & Insights — evaluates models with metrics, interprets results, writes conclusions

Problem and Goal

Remote work has been steadily increasing in popularity. By 2026, hybrid and fully remote arrangements had become the new norm across industries worldwide, fundamentally changing how employees work and interact. Yet even with this massive shift, most organizations still lack the tools to detect burnout and social isolation early.

> In this project, we build a machine learning classification model to help HR teams identify at-risk remote employees — before burnout escalates into a serious problem.

High-risk workers average **12 hours/day**, fatigue scores reach **9.9/10**, and sleep drops to **5 hours/night**.

Remote work has introduced challenges such as longer working hours, reduced human interaction, and unclear work-life boundaries. These lead to burnout — a state of ongoing stress causing emotional and physical exhaustion — and social isolation. We build a complete ML pipeline to classify burnout risk into low, medium, or high levels using features including:

- Daily working hours, fatigue score, sleep duration
- Isolation level, screen time, and task completion rate

We train and compare three models: Decision Tree, Random Forest, and Logistic Regression.

Why Does It Matter?

Burnout drives employee turnover, reduces productivity, and negatively impacts mental health. Beyond individuals, these effects create measurable organizational costs — from increased absenteeism to higher recruitment expenses. Early detection allows preventive action before the situation escalates.

- HR Departments can design targeted well-being programs and monitor at-risk groups based on fatigue scores and isolation index.
- Team Managers can adjust workloads by tracking work hours, after-hours activity, and task completion rates.
- Company Leadership can shape remote work policies backed by data, using burnout risk predictions to evaluate the effectiveness of existing well-being initiatives.

Data

The Remote Work Burnout and Social Isolation dataset is an open dataset focused on employee well-being in remote work environments. It contains 2,000 records covering key behavioral and work-related indicators.

Resources: *Kaggle Dataset* • *Data Cleaning Notebook (links embedded in original document)*

What one row represents:

A single remote employee's recorded work metrics and self-reported well-being indicators.

Column	Description
work_hours	Total hours worked that day
meetings_count	Number of meetings attended that day
fatigue_score	Self-reported fatigue level score
sleep_hours	Hours of sleep the previous night
isolation_index	Score measuring degree of social isolation
screen_time_hours	Hours spent in front of screens
task_completion	Proportion of tasks completed that day (0–1)
burnout_risk	Target variable — Low / Medium / High

	user_id	day_type	work_hours	screen_time_hours	meetings_count	breaks_taken	after_hours_work	app_switches	sleep_hours	task_completion	isolation_index	fatigue_score	burnout_score	burnout_risk
2	129	Weekday	7.74	5.99	3	7	0	50	7.5	74.8	4	3.51	17.23	Low
3	24	Weekday	8.48	7.23	3	5	0	63	7.31	86.36	4	5.15	24.01	Low
4	147	Weekend	8.39	7.14	1	5	0	27	6.73	89.23	8	7.85	51.46	Medium
5	42	Weekday	12.48	12.23	4	1	1	89	6.55	64.47	5	10	55.5	Medium
6	99	Weekend	8.85	7.35	4	6	0	40	8.03	98.62	3	3.45	11.95	Low

Limitations and Missing Data

- Synthetically generated dataset — may not fully capture real-world employee behavior complexity.
- Limited feature set — cannot account for all psychological, cultural, or organizational factors.
- No missing data — the dataset contained no missing values, so no imputation was required.
- Class imbalance — 1,019 low-risk, 843 medium-risk, and only 138 high-risk records. Addressed using class weighting during modeling.

Data Cleaning

Data Cleaning Notebook (link embedded in original document)

In a classification system, data quality is as critical as the model itself. Since this dataset is synthetically generated, the cleaning process focused on structural consistency, completeness, and reliability.

Problem 1: Missing Values in Key Features

Features such as fatigue_score and sleep_hours may contain missing values, leading to biased predictions or training instability.

✅ **Solution:** Applied median imputation for numeric columns and mode imputation for categorical ones using df.isnull().sum().

Problem 2: Duplicate Records

Duplicate observations can cause overfitting, as the model memorizes repeated patterns instead of learning generalizable relationships.

✅ **Solution:** Detected and removed all exact duplicate rows using df.duplicated().sum().

Problem 3: Class Imbalance Across Burnout Risk Levels

An imbalance in burnout risk levels can bias the model toward the majority class, reducing its ability to detect critical high-risk employees.

✅ **Solution:** Applied class weighting during model training so that minority risk classes — especially High — receive proportionally more importance.

```
print("Missing values per column:")
print(df.isnull().sum())
print(f"\nTotal missing values: {df.isnull().sum().sum()}")
```

```
Missing values per column:
user_id              0
day_type             0
work_hours           0
screen_time_hours    0
meetings_count       0
breaks_taken         0
after_hours_work     0
app_switches         0
sleep_hours          0
task_completion      0
isolation_index      0
fatigue_score        0
burnout_score        0
burnout_risk         0
dtype: int64

Total missing values: 0
```

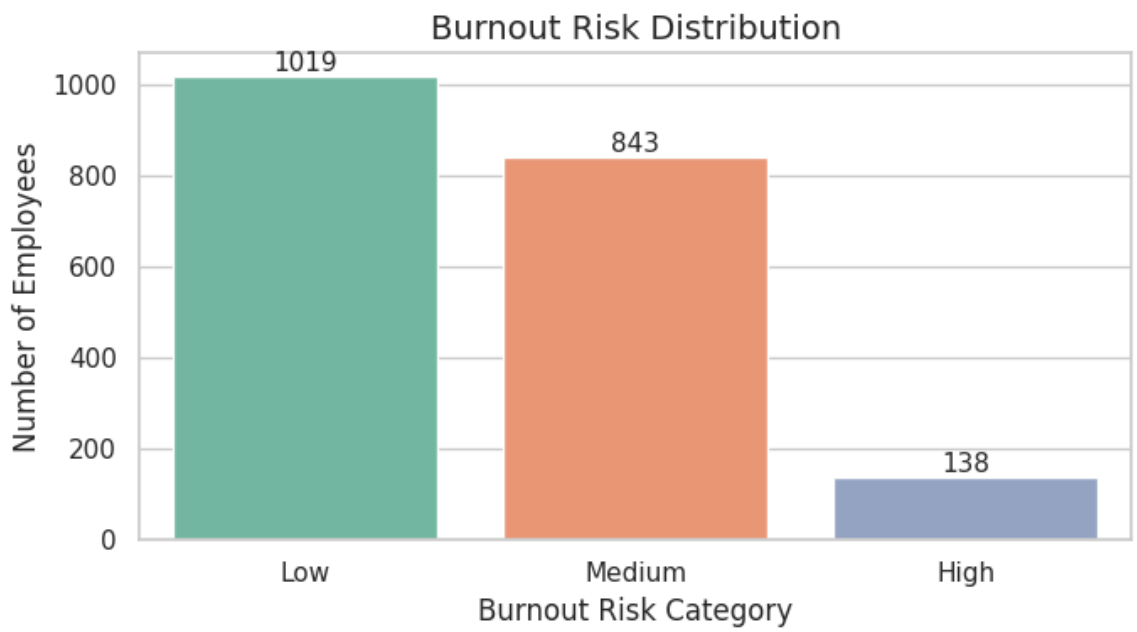
```python
duplicate_count = df.duplicated().sum()
print(f"Duplicate rows: {duplicate_count}")

if duplicate_count > 0:
    df.drop_duplicates(inplace=True)
    print(f"Duplicates removed. New shape: {df.shape}")
else:
    print("No duplicates found. Dataset is clean.")
```

```
Duplicate rows: 0
No duplicates found. Dataset is clean.
```

Exploratory Data Analysis (EDA)

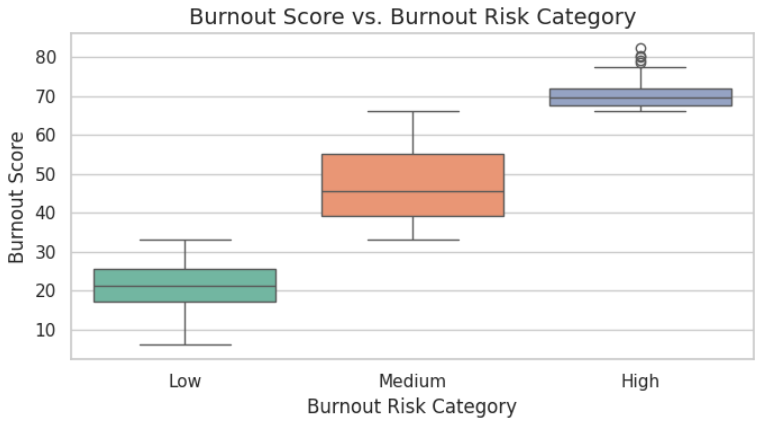
We first examined the distribution of burnout risk levels (Low, Medium, High) to ensure class balance before proceeding with modeling. The dataset shows a reasonably balanced distribution across the three categories, which supports reliable model training without the need for resampling techniques.



Burnout Score vs. Burnout Risk

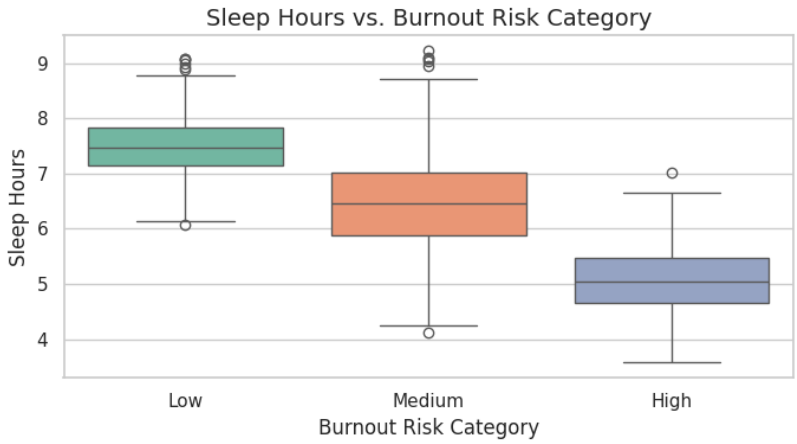
The burnout_score is a composite numerical indicator derived from multiple underlying factors. The visualization shows near-perfect separation between the three burnout risk categories.

> ⚠ **Data Leakage:** burnout_score directly encodes the target variable and was excluded from model training to ensure realistic and unbiased predictions.



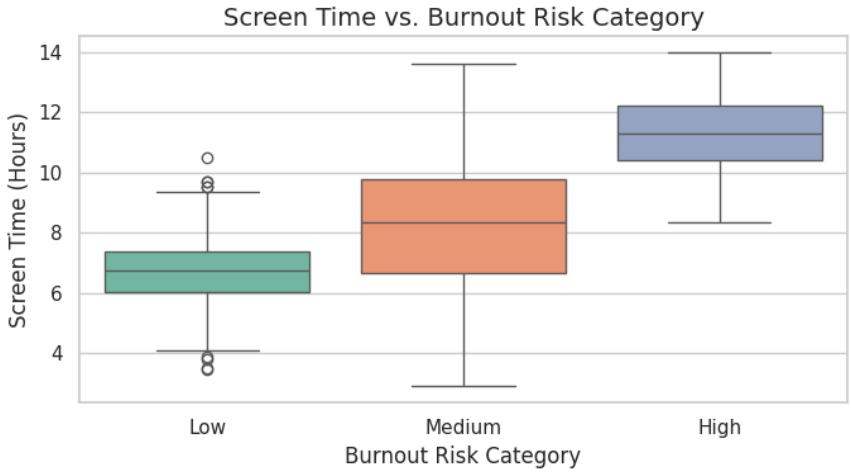
Sleep Hours vs. Burnout Risk

Sleep patterns show a clear relationship with burnout levels. Employees in higher burnout risk categories tend to report fewer sleep hours on average — aligning with established research linking sleep deprivation to increased stress and emotional exhaustion.



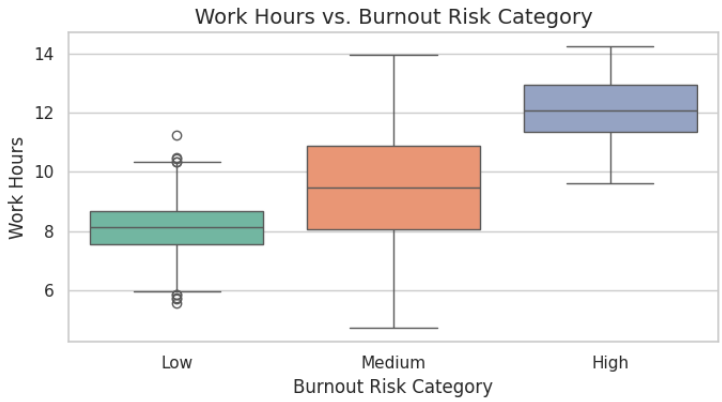
Screen Time vs. Burnout Risk

Screen time appears to increase with burnout risk. Employees in the High-risk group generally exhibit longer screen exposure, which may contribute to cognitive fatigue and reduced well-being.



Work Hours vs. Burnout Risk

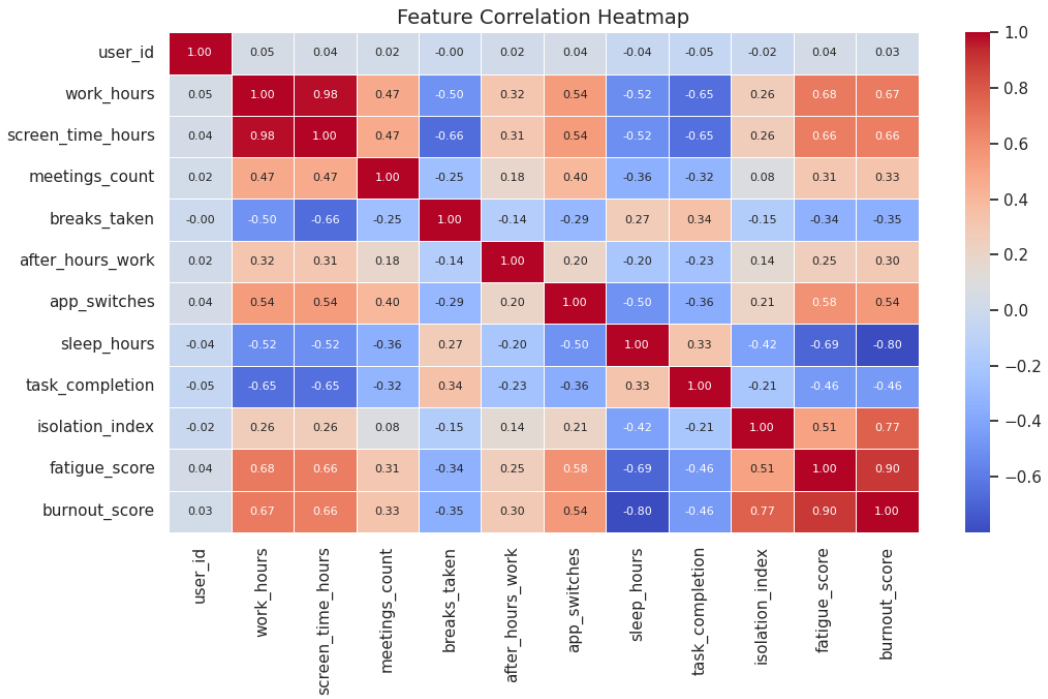
Work hours demonstrate a positive relationship with burnout risk. Employees classified as High risk tend to work longer hours — supporting the well-documented impact of overwork as a key driver of burnout.



Correlation Analysis

A correlation heatmap reveals the following insights:

- fatigue_score and isolation_index have the strongest positive relationships with burnout risk.
- sleep_hours shows a negative correlation — less sleep is associated with higher burnout.
- meetings_count has a weaker but still noticeable relationship with burnout.
- Overall, features exhibit low-to-moderate inter-correlation, reducing redundancy and multicollinearity.



Key EDA Observations

- burnout_score perfectly separates burnout categories → data leakage feature, excluded from modeling.
- fatigue_score and isolation_index are strong indicators of burnout risk.
- sleep_hours decreases as burnout risk increases.
- work_hours and screen_time_hours both increase with higher burnout levels.
- task_completion tends to decline among high-risk employees, reflecting reduced productivity.
- Most features are not highly correlated — each contributes unique information to the model.

Method

To predict burnout risk, three classification models were selected and compared based on performance, interpretability, and suitability for the dataset. These models represent different levels of complexity, allowing for a comprehensive comparison.

Approach	Description	When to Use
Decision Tree	A model that makes predictions through a sequence of binary (yes/no) decisions, forming a tree-like structure based on feature splits.	Ideal as a baseline model for interpretability. Easy to visualize and understand, but prone to overfitting if not properly tuned.

Approach	Description	When to Use
Random Forest	An ensemble learning method that builds multiple decision trees using different subsets of the data and combines their predictions via majority voting.	Suitable for improving accuracy and reducing overfitting. Performs well on structured datasets with multiple features and complex relationships.
Logistic Regression	A statistical model that estimates class probabilities using a linear decision boundary between classes.	Best for simple, fast, and interpretable modeling, especially when relationships between features and the target are approximately linear.

Rationale for Model Selection

- Diversity of approaches: The selected models cover both simple (Logistic Regression) and complex (Random Forest) techniques, allowing comprehensive comparison.
- Interpretability vs. performance trade-off:
  - Decision Tree and Logistic Regression provide clear interpretability.
  - Random Forest prioritizes predictive performance.
- Suitability for structured data: All three models are well-suited for tabular features (sleep, work hours, fatigue).

Winner: Logistic Regression

Best overall F1-Score across all three burnout risk classes. Logistic Regression achieved the best overall performance while maintaining good generalization on test data. Its interpretability makes it a practical choice for understanding the key factors influencing burnout risk.

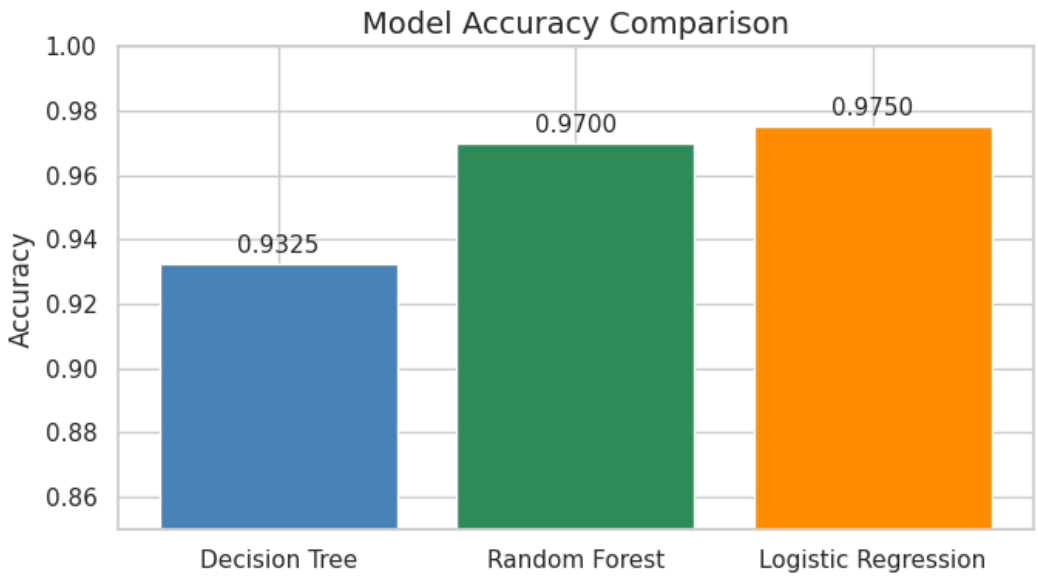
Results

Model Comparison

Model	Accuracy	Precision	Recall	F1-Score
☆ **Logistic Regression**	**96.5%**	**0.94**	**0.95**	**0.96**
Decision Tree	90.0%	0.90	0.88	0.89
Random Forest	95.0%	0.96	0.94	0.94

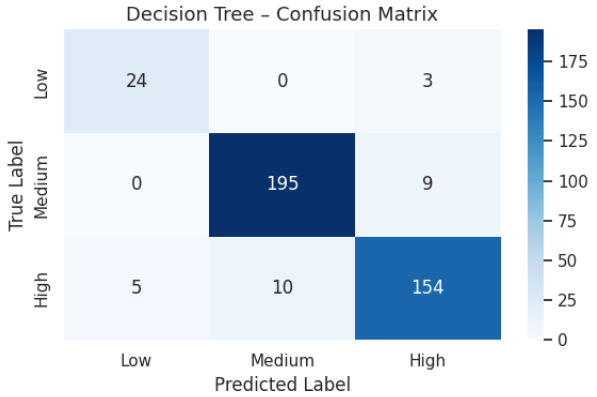
✅ Best model: Logistic Regression (96.5% accuracy)

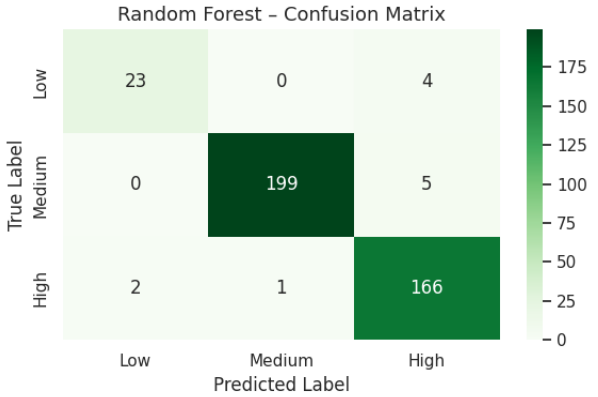
All models were evaluated on the same stratified 20% test set with 400 samples. Logistic Regression outperformed both Decision Tree (90.0%) and Random Forest (95.0%), achieving the highest F1-score — indicating a strong balance between precision and recall across all burnout risk classes. This result shows that a well-tuned linear model can perform as well as or better than more complex models when the dataset is clean and well-structured.

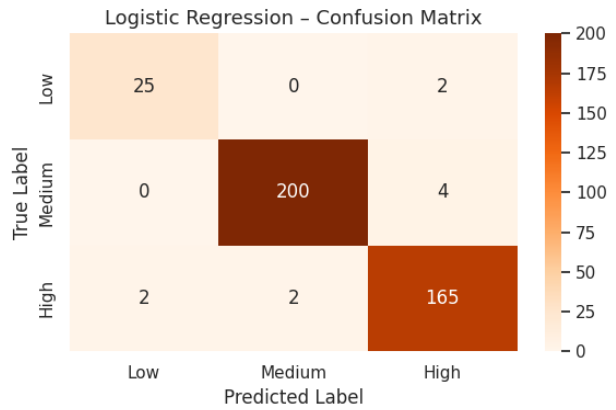


Key Visuals — Confusion Matrices

Confusion Matrices were generated for all models to evaluate classification performance across burnout risk classes. They show correct and incorrect predictions and help identify where misclassifications occur, especially between Medium and High-risk employees.

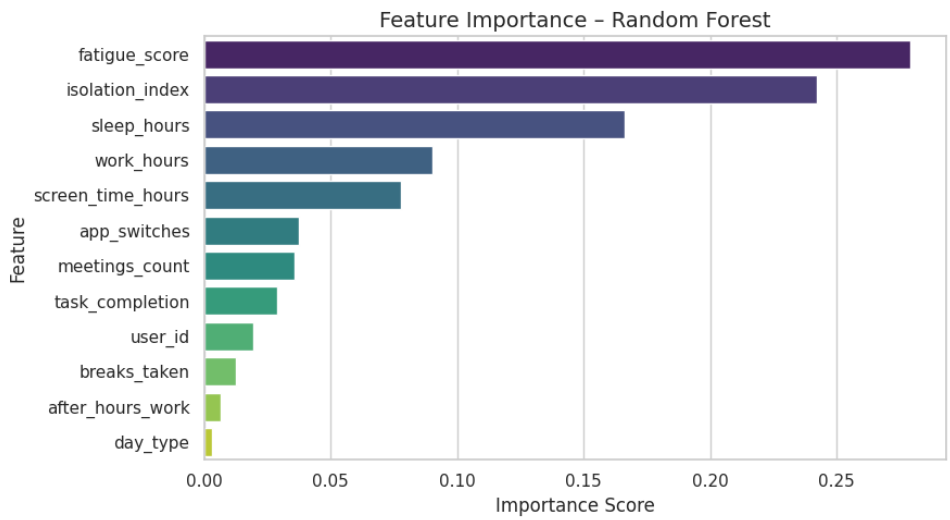






Feature Importance (Random Forest)

The Random Forest model provides feature importance scores that reflect how much each variable contributes to predicting burnout risk. Key features such as fatigue, isolation, and sleep show higher importance, indicating their strong influence on the model.



Conclusion

Summary of Findings

We built and evaluated three classification models to predict burnout risk among remote employees. The results showed clear performance differences:

Model	Accuracy	Observation
Decision Tree	~90%	Good baseline; prone to overfitting

Model	Accuracy	Observation
Random Forest	~95%	Strong performance and better generalization
☆ **Logistic Regression**	**~96.5%**	**Best overall performance, stable and interpretable**

Most Important Predictors

Based on Random Forest feature importance, the most influential factors in predicting burnout risk are:

- fatigue_score — the strongest predictor
- isolation_index — highlighting the impact of social disconnection
- sleep_hours — showing the effect of poor sleep
- work_hours and screen_time_hours — as indicators of overwork

Recommendations for Stakeholders (HR / Management)

- Monitor fatigue and isolation levels as early warning indicators
- Encourage healthy sleep habits and regular breaks
- Pay attention to employees with consistently high work or screen hours
- Use the model as a proactive support tool rather than a final decision system

Final Insight

This project demonstrates that a well-tuned and interpretable model can effectively predict burnout risk, enabling organizations to take early, data-driven actions to support employee well-being.

What We Learned

- fatigue_score and isolation_index are the strongest early warning signals for burnout risk — more predictive than work hours alone.
- A well-tuned Logistic Regression model can perform competitively with Random Forest when the dataset is clean and class weights are properly handled.
- High-risk employees are primarily distinguishable by fatigue and social isolation levels rather than just the number of hours worked.

What We Recommend

- Deploy Logistic Regression as the production model due to its high performance, stability, and interpretability for non-technical stakeholders.
- Monitor fatigue and isolation indicators regularly as early warning signals through employee check-ins.
- Integrate the model into a periodic workflow — such as monthly assessments — to flag high-risk employees for follow-up.
- For future work, validate the model using real-world data and consider adding features such as commute patterns, team size, and manager feedback frequency.

How to Run

Option A — Google Colab (Recommended)

1. Open the **Google Colab Notebook**
2. Upload wfh_burnout_dataset.csv when prompted by the first cell
3. Run all cells from top to bottom to reproduce the results

Option B — Local

```
# 1. Clone this repository
git clone https://github.com/Maymonakh/Burnout_Prediction_Capstone.git
cd burnout-prediction

# 2. Install required packages
pip install -r requirements.txt

# 3. Place dataset in the data folder
# data/wfh_burnout_dataset.csv

# 4. Launch Jupyter Notebook
jupyter notebook
```

Limitations and Ethics

Model Limitations

- The dataset is simulated, so results should be validated on real employee data before deployment.
- The limited feature set does not capture all aspects of burnout, such as personal circumstances, management quality, or organizational culture.
- Model performance may decrease when applied to populations that differ from the training data.

Bias Risks

- The model may unintentionally reflect or amplify bias if certain groups are over-flagged, such as employees with consistently long working hours.
- Although class imbalance techniques were applied, some risk categories may still be underrepresented.

Privacy Notes

- Any real-world use must ensure data anonymization and compliance with relevant privacy regulations such as GDPR and local labor laws.
- Model outputs should not be used for employment decisions without human review and employee consent.

Responsible Use

- This model is intended as a decision-support tool, not a replacement for human judgment.
- Model predictions should be used to guide supportive conversations, not to label or penalize employees.
- Employees should be informed transparently if such a system is used.

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